AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

MAY 2011

Physics Paper II

Time allowed: 2 hours 25 minutes Marks 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL NINE questions. Questions 7, 8 & 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

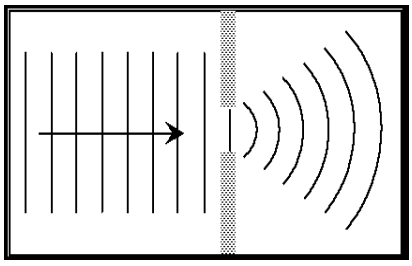
   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

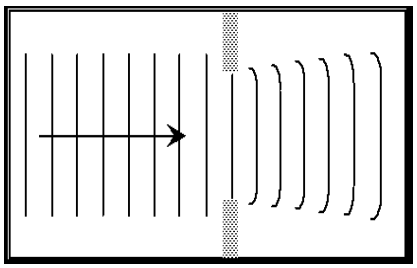
5. You may use a simple calculator if you wish.

Q.1. (Total 4 Marks)

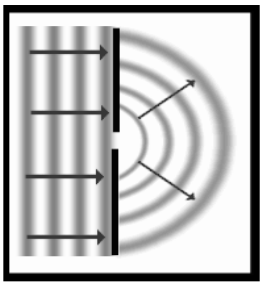
Name the property of the wave which is shown in the figures below and describe briefly what is happening in each figure.



(A)



(B)



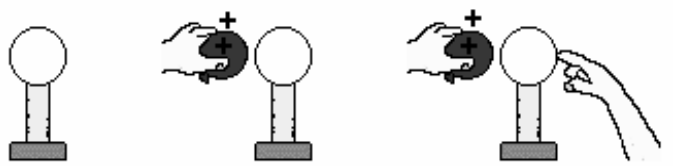
(C)

Q.2. (Total 5 Marks)

A convex lens is placed between an object and the screen to get a real image which is magnified 3 times. If the distance between the object and the screen is 60 cm, find the value of p and q.

Q.3. (Total 6 Marks)

a. A positively charged balloon is brought near a neutral sphere as shown in the given figure. While the balloon is near the sphere, it is touched to the ground.



(A) (B) (C)

i. Name the phenomenon which is shown in the above figure. (1 Mark)

_____

_____

ii. Show distribution of charges on the figure B. (1 Mark)

_____

_____

iii. In which direction do the charges flow in the figure C? (1 Mark)

_____

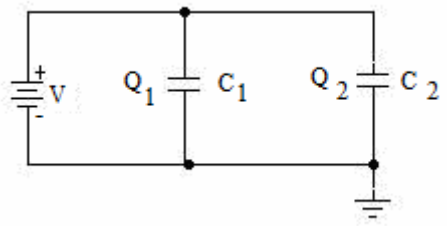
_____

iv. Define parallel plate capacitor. (1 Mark)

_____

_____

b. Find the equivalent capacitance of two capacitors connected in parallel. (2 Marks)



Q.4. (Total 3 Marks)

Electromagnets are the devices consisting of a solenoid. An electric current passed through the coil induces a magnetic field. When the iron core is placed in this field, microscopic domains that can be considered small permanent magnets in the iron align themselves in the direction of the field, thus increasing greatly the strength of the magnetic field produced by the solenoid. The magnetization of the core reaches saturation once all the domains are completely aligned. When the current is switched off, the core retains only a week residual magnetism.

i. Define electromagnetism. (1 Mark)

_____

_____

ii. Why is iron core placed in the solenoid? (1 Mark)

_____

_____

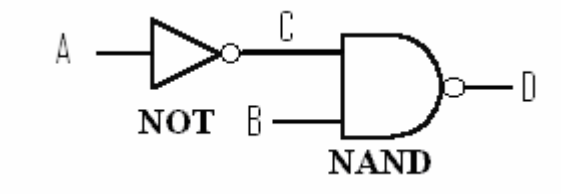
iii. If all domains align themselves, then what state will be attained? (1 Mark)

_____

_____

Q.5. (Total 3 Marks)

Write a truth table for the following combination of gates.



Q.6. (Total 4 Marks)

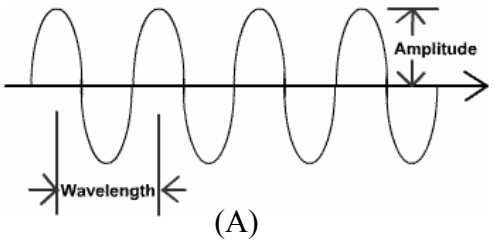
Write any FOUR points of difference among α – particles, β – particles and γ – rays.

α – particles	β – particles	γ – rays

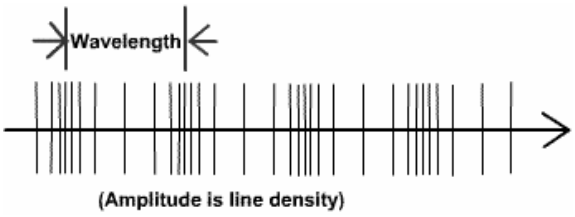
Q.7. (Total 5 Marks)

EITHER

a. Two types of wave are shown in the figures given below. Identify and define each type of wave and explain any ONE of them.



(A)



(B)

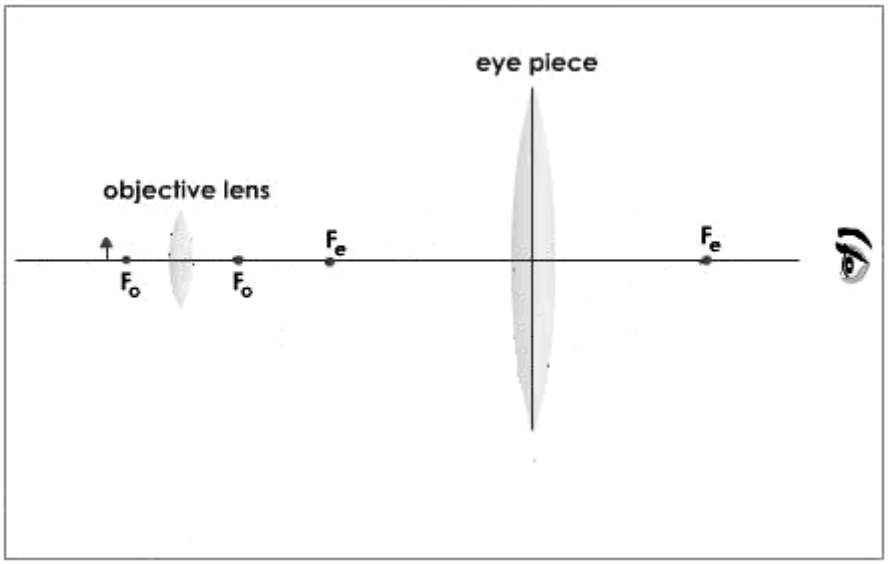
OR

b. Describe any FIVE uses of ultrasound in our daily life.

Q.8. (Total 5 Marks)

EITHER

a. Complete the given ray diagram of a compound microscope.



OR

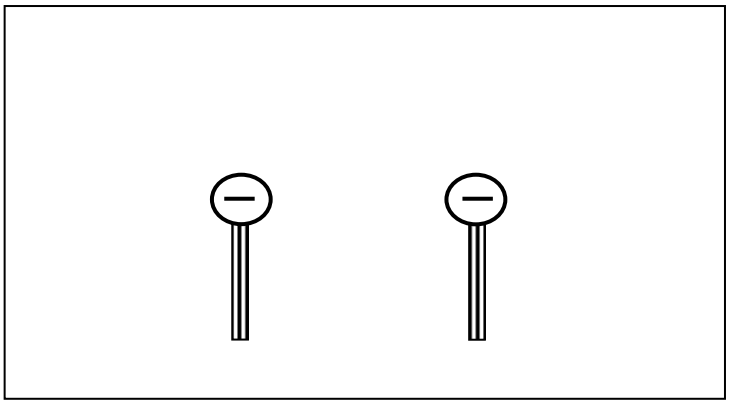
b. Where should an object be placed in front of a concave mirror of focal length 12 cm, so as to get its real image magnified 4 times?

Q.9. (Total 5 Marks)

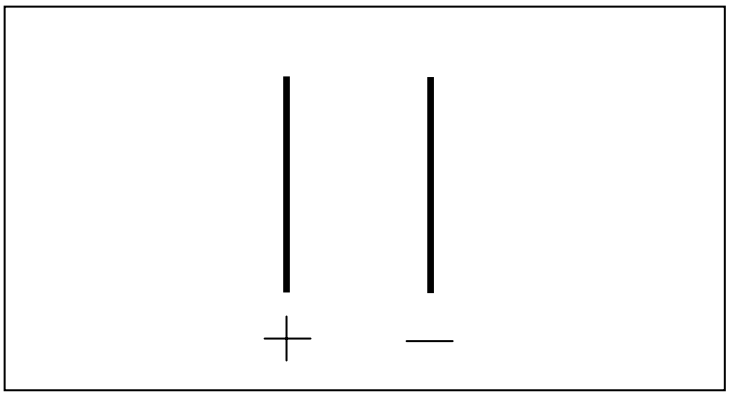
EITHER

a. Draw an electric field pattern in each of the following diagram.

i. Two negatively charged brass spheres are mounted on insulation rods



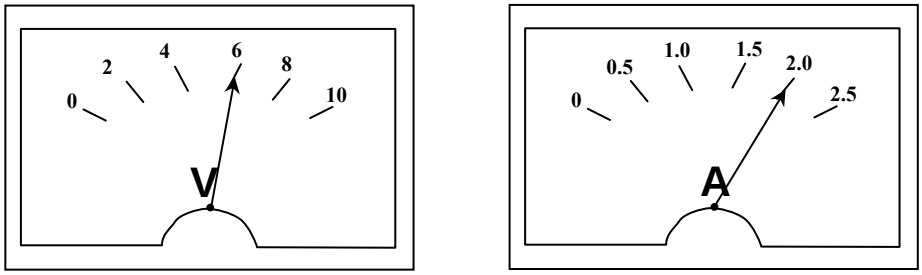
ii. Two oppositely charged ebonite plates



Plastics and clothes are insulators but when they are rubbed with each other, static electricity is produced. How does this happens?

OR

b. The given diagrams show the readings of an electric current through a resistor and the potential difference across it.



Calculate the power dissipated in the resistor and the energy used in units when 3 kW room heater is turn on for 4 hours.

END OF PAPER

Please use this page for rough work